

ANKIT KUMAR

Department of Management Sciences (DoMS), Indian Institute of Technology Kanpur, India
ankitk25@iitk.ac.in • [LinkedIn: in/ankit-kumar](#) • [GitHub: Ankitiitk369](#) • [Portfolio: ankit369portfolio](#)

Education

Indian Institute of Technology, Kanpur

Master of Technology (M.Tech), Department of Management Sciences (DoMS)

Kanpur, India
July 2025 – Present • CPI: **7.18 / 10.0**

C.S.A. University of Agriculture and Technology

Bachelor of Technology (B.Tech) in Agricultural Engineering Aug 2021 – June 2025 • CPI: **8.25 / 10.0** (*Chancellor's Silver Medalist*)
Kanpur, India

Professional Experience

Tata Consultancy Services (TCS)

Data Science & AI Intern | Computer Vision & Machine Learning

Remote / India
May 2026 – July 2026

- Modeled multi-vehicle trip trajectories using Multi-Origin Convergence Routing with Haversine & OSRM matrices for fleet carbon density.
- Built an automated multi-sheet Excel reporting pipeline and interactive **Streamlit BI Dashboard** with GIS maps and **17-test Pytest** suite.
- Trained dual-stage ensemble estimators: Gradient Boosting ($R^2 = 0.996$, RMSE = **3.12** g/km) and Random Forest.

Key Flagship Projects: SQL Analytics, Power BI & Business Intelligence

Customer Shopping Behavior Analysis & BI Platform

Sep 2025 – Oct 2025

Stack: PostgreSQL, Power BI, Advanced DAX, Pandas, Relational Modeling

[\[GitHub Repo\]](#)

- Analyzed **3,900 consumer purchase records** across 18 demographic and transactional dimensions using Python and relational modeling.
- Formulated complex **PostgreSQL** queries using **CTEs**, window ranking functions (`ROW_NUMBER`, `DENSE_RANK`), and conditional joins.
- Engineered an executive **Power BI** dashboard with dynamic **DAX measures**, RFM segmentation, and drill-through cohort analysis.

Loan Portfolio Risk Analysis & Visualization (Tableau)

Sep 2025 – Oct 2025

Stack: Tableau Desktop, SQL, Financial KPIs, Risk Modeling

[\[GitHub Repo\]](#)

- Appraised **38,580 loan records** to examine funded capital, repayment schedules, interest distributions, and default risks.
- Built **3 production-grade Tableau executive dashboards** tracking DTI ratios, monthly delinquencies, and capital at risk.

Zepto Quick-Commerce SQL Data Analysis

Aug 2025 – Sep 2025

Stack: PostgreSQL, Advanced SQL, EDA, Dark Store Logistics

[\[GitHub Repo\]](#)

- Evaluated 3,700+ product catalog records across 14 categories; computed category GMV, average order values (AOV), and stockout frequency.
- Formulated dark store inventory reorder thresholds to prevent high-velocity SKU stockouts during peak ordering hours via SQL CTEs.

Additional Relevant Analytics, Econometrics & BI Projects (Full Portfolio)

- BigMart Sales Multivariate Regression:** Econometric diagnostics (Breusch-Pagan, Jarque-Bera, VIF) for retail outlet revenue drivers.
- Marketing Campaign Performance Analytics:** A/B hypothesis testing, Two-Way ANOVA, Chi-Square, and CAC acquisition modeling.
- McDonald's Customer Perception Segmentation:** Unsupervised K-Means clustering and PCA perceptual mapping across QSR brand perception.
- Retail Mall Shoppers Clustering Analysis:** Evaluated K-Means, Hierarchical, and DBSCAN clustering with Silhouette score validation.
- Netflix Titles Catalog Exploratory Analytics:** Global content release trends, movie vs. TV show ratios using Seaborn and Plotly.
- Deloitte Australia Virtual Analytics Simulation:** Forensic KPI analytics, business requirements synthesis, operational reporting.
- VitalSync AI Environmental & Health IoT:** Correlating outdoor Air Quality Index (AQI) with aerobic athletic performance telemetry.

Technical Skills, Coursework & Certifications

- BI & Analytics Stack:** PostgreSQL, MySQL, Advanced SQL (CTEs, Window Functions), Power BI (DAX), Tableau, Advanced Excel, Python
- Statistical Methods & Tooling:** Pandas, NumPy, SciPy, Statsmodels, A/B Testing, ANOVA, OLS Regression, Time Series, Streamlit, Git
- Relevant Courses (IIT Kanpur):** Statistical Modelling for Business Analytics, Operations Research, Data Mining, Probability & Statistics
- Honors & Certifications:** GATE 2025: **98.48 %ile** • **Chancellor's Silver Medal** • **SEBI-Investor Certification** (NISM) • **Deloitte Australia**